

ASHUTOSH DWIVEDI

AI / ML Engineer | NLP · Scikit-learn · LLM Apps · Model Deployment

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AI-focused engineering student with hands-on experience across the ML lifecycle — data collection, feature engineering, training and deployment. Built NLP/GenAI systems using spaCy, Whisper, Transformers and OpenAI/Ollama LLMs. Seeking an AI/ML Engineer (or AI intern) role to turn research quality models into working products.

EDUCATION

B.Tech — Computer Science & Engineering

2023 – 2026 (Expected)

United Institute of Technology, Prayagraj (United University)

TECHNICAL SKILLS

Machine Learning	scikit-learn, Random Forest, classification, feature engineering, model evaluation
NLP & GenAI	spaCy NER, Hugging Face Transformers, Whisper (STT), OpenAI API, Ollama (local LLMs)
Data Handling	pandas, NumPy, text normalisation, feature extraction
Deployment	Flask, FastAPI, Streamlit, model serialisation (joblib/pickle)
Languages & Tools	Python, SQL, Git/GitHub, VS Code, Linux

PROJECTS

Voice-to-Prescription — AI Medical Scribe

Tech: Python, FastAPI, Whisper (STT), spaCy NER, Ollama LLM, Streamlit

- Built an end-to-end voice-to-prescription pipeline that converts a doctor's spoken dictation into structured, validated JSON prescriptions — designed to run fully local, with no cloud dependency.
- Integrated Whisper speech-to-text with a spaCy NER + rule-based engine to normalise medicine names, dosages, frequencies and durations; added optional Ollama (Mistral 3B) LLM for higher-accuracy extraction.
- Exposed FastAPI REST endpoints (CORS-enabled) for mobile clients and a Streamlit demo UI with field-level validation and confidence scores.

AI-Powered Phishing Detection System

Tech: Python, Flask, scikit-learn (Random Forest), JavaScript, HTML, CSS

- Developed a web app that classifies URLs as phishing / legitimate in real time using a Random Forest model trained on lexical URL features with scikit-learn.
- Built the complete stack end-to-end — model training pipeline, Flask REST backend and a responsive cybersecurity-themed frontend (HTML/CSS/JS).

ID-Card OCR & Information Extraction

Tech: Python, OpenCV, pytesseract, Pillow, Streamlit, Flask

- Engineered an OCR pipeline that extracts structured fields (name, DOB, ID numbers) from Aadhaar, PAN, driving licence and voter ID images.
- Implemented dedicated per-document extractors with OpenCV/Pillow image pre-processing for higher recognition accuracy; exposed via Streamlit app and Flask API.

AI Text Summarization Web App

Tech: Python, Flask, JavaScript, HTML, CSS

- Built a web app that condenses long text into concise summaries through a clean two-pane interface with instant Flask-powered processing.

ACHIEVEMENTS & CERTIFICATIONS

- Competed at Rajasthan Police Hackathon 1.0 (RAKAM), Jaipur (Jan 2024)
- Represented Team BLACKBOX 913 at HackQuest '25 — 24-hr hackathon, UIT Prayagraj (Apr 2025)
- Team Cryptoknights at HackDiwas 2024 — 24-hr hackathon, United University (Apr 2024)